

Table 2: Overview of the diversity scores for the top 20 news recommendations and AUC scores for predicting user impressions for the Microsoft News Dataset (MIND). MIND covers predominantly news from the US. We therefore model our distribution with the Democratic and Republican Party as majority and the rest as minority. NTD has buckets for articles that mention 1) the Democratic Party (15%), 2) the Republican Party (15%), 3) the Democratic *and* Republican Party (15%), 4) minority parties (15%, *must* mention at least one minority party, *may* mention a majority party), and 5) no political party (40%). The effective distribution in MIND of political party mentions is 3% for the Democratic Party, 4% for the Republican Party, 1% for the Democratic and Republican Party, 6% for other parties, and 86% of articles have no mention. The values closest to the perfect score are highlighted in **green, second closest in **blue**, and third closest in **red**.**

Model	Re-ranking	Activ.	Cat. Calib.	Comp. Calib.	Frag.	Alt. Voices	Repr.	Cat. Gini	Sent. Gini	Party Gini	Cat. ILD	Sent. ILD	Party ILD	Train. Cost	Rec. Cost	AUC
Perfect (NTD)	0.000	1.000	1.000	0.000	0.000	0.000	1.000	0.000	0.133	0.250	1.000	0.779	0.789	N/A	N/A	1.000
LSTUR [1]	N/A	0.266	0.620	0.313	0.503	0.051	0.296	0.855	0.593	0.905	0.618	0.585	0.247	625	567	0.593
NPA [43]	N/A	0.191	0.574	0.329	0.443	0.076	0.280	0.792	0.614	0.854	0.751	0.587	0.323	1205	352	0.595
NRMS [44]	N/A	0.211	0.559	0.313	0.712	0.071	0.305	0.756	0.554	0.896	0.764	0.615	0.246	645	953	0.626
LSTUR + NTD	G-KL	0.250	0.619	0.311	0.558	0.097	0.408	0.786	0.133	0.250	0.769	0.779	0.789	625	61	N/A
	PM-2	0.227	0.607	0.317	0.574	0.086	0.418	0.809	0.133	0.250	0.734	0.779	0.789	625	53	N/A
	MMR	0.287	0.604	0.334	0.594	0.102	0.471	0.766	0.000	0.000	0.781	0.789	0.842	625	126	N/A
NPA + NTD	G-KL	0.226	0.585	0.327	0.458	0.098	0.434	0.829	0.133	0.250	0.691	0.779	0.789	1205	61	N/A
	PM-2	0.251	0.599	0.331	0.465	0.117	0.438	0.858	0.133	0.250	0.651	0.779	0.789	1205	53	N/A
	MMR	0.320	0.581	0.329	0.475	0.137	0.517	0.837	0.000	0.000	0.691	0.789	0.842	1205	126	N/A
NRMS + NTD	G-KL	0.232	0.563	0.315	0.692	0.075	0.402	0.713	0.133	0.250	0.812	0.779	0.789	645	61	N/A
	PM-2	0.238	0.560	0.311	0.706	0.079	0.390	0.729	0.133	0.250	0.801	0.779	0.789	645	53	N/A
	MMR	0.284	0.574	0.322	0.692	0.088	0.468	0.716	0.000	0.000	0.809	0.789	0.842	645	126	N/A
D-RDW	N/A	0.215	0.612	0.297	0.455	0.110	0.481	0.711	0.133	0.250	0.836	0.779	0.789	69	13	0.486
RP^3_β [11]	N/A	0.210	0.555	0.310	0.706	0.071	0.308	0.726	0.544	0.897	0.806	0.621	0.245	69	2	0.504
RWE-D [27]	N/A	0.114	0.587	0.298	0.378	0.108	0.389	0.651	0.553	0.758	0.878	0.629	0.459	78	1	0.488
Random	N/A	0.197	0.618	0.314	0.702	0.057	0.301	0.663	0.503	0.897	0.861	0.649	0.251	N/A	N/A	0.498